

Python Exception Handling

Complete Python code covering try-except, multiple exceptions, else, finally, raise, and common exceptions.

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# =====
# PYTHON EXCEPTION HANDLING
# =====

# 1. Basic try-except
print("1. Basic try-except")

try:
    a = 10
    b = 0
    print(a / b)

except ZeroDivisionError:
    print("Cannot divide by zero")

# =====
# 2. Specific Exception
# =====
print("\n2. Specific Exception")

try:
    num = int("abc")
    print(num)

except ValueError:
    print("Invalid value")

# =====
# 3. Multiple except Blocks
# =====
print("\n3. Multiple except Blocks")

try:
    num = int(input("Enter a number: "))
    result = 100 / num
    print("Result:", result)

except ValueError:
    print("Please enter a valid number")

except ZeroDivisionError:
    print("Cannot divide by zero")

# =====
# 4. else Block
# =====
print("\n4. else Block")

try:
    a = 10
    b = 2
    result = a / b

except ZeroDivisionError:
    print("Cannot divide by zero")

else:
    print("Division successful")
    print("Result:", result)

# =====
# 5. finally Block
# =====
print("\n5. finally Block")

try:
    a = 10
    b = 0
    print(a / b)

except ZeroDivisionError:
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    print("Error occurred")

finally:
    print("This block always executes")

# =====
# 6. Getting the Error Message using 'as'
# =====
print("\n6. Error Message")

try:
    print(10 / 0)

except ZeroDivisionError as e:
    print("Error:", e)

# =====
# 7. IndexError
# =====
print("\n7. IndexError")

numbers = [10, 20, 30]

try:
    print(numbers[5])

except IndexError:
    print("Index does not exist")

# =====
# 8. KeyError
# =====
print("\n8. KeyError")

student = {
    "name": "Pratiksha",
    "age": 20
}

try:
    print(student["marks"])

except KeyError:
    print("Key does not exist")

# =====
# 9. TypeError
# =====
print("\n9. TypeError")

try:
    result = 10 + "20"
    print(result)

except TypeError:
    print("Cannot add integer and string")

# =====
# 10. FileNotFoundError
# =====
print("\n10. FileNotFoundError")

try:
    file = open("abc.txt", "r")
    print(file.read())
    file.close()

except FileNotFoundError:
    print("File does not exist")

# =====
# 11. raise Keyword
# =====
print("\n11. raise Keyword")

age = 15

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try:
    if age < 18:
        raise ValueError("Age must be 18 or above")

    print("Eligible")

except ValueError as e:
    print("Error:", e)

# =====
# 12. Complete try-except-else-finally
# =====
print("\n12. Complete Example")

try:
    num = int(input("Enter a number: "))
    result = 100 / num

except ValueError:
    print("Invalid input")

except ZeroDivisionError:
    print("Cannot divide by zero")

else:
    print("Calculation successful")
    print("Result:", result)

finally:
    print("Program completed")

# =====
# IMPORTANT NOTES
# =====

# try      -> Code that may cause an error
# except   -> Handles the error
# else     -> Runs when there is no error
# finally  -> Always runs
# raise    -> Manually creates an exception

print("\n=====")
print("Exception Handling Program Completed")
print("=====")

```